



Criminal courts' artificial intelligence: the way it reinforces bias and discrimination

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Abstract

Embracing, pervasive, and unstoppable global algorithmization greatly influences the deployment of artificial intelligence systems in criminal courts to replace obsolete bail and sentencing practices, reduce recidivism risk, and modernize judicial practices. Since artificial intelligence systems have provably appeared to have the duality of golden promises and potential perils, applying such a system in the justice system also entails some associated risks. Hence, allocating this unchecked-novel resource in judicial domains sparks vigorous debate over their legal and ethical implications. With such backgrounds, this paper examines how and why artificial intelligence systems reinforce bias and discrimination in society and suggest what approach could be an alternative to the current predictive justice mechanisms in use.

Keywords Artificial intelligence · Risk prediction · Machine bias · Predictive justice

1 Introduction

Rapid developments in probabilistic computational power, big data, and machine learning are notably influencing the deployment of artificial intelligence (AI) systems¹ in public bodies with a view to outsourcing authority to machines in performing public functions, such as social welfare, law enforcement, prison, and administration of justice. So far, as criminal justice is concerned, the United States of America is predominantly at the forefront in embracing AI tools in such a public sphere, considering as a part of rampant algorithmization or “algorithmic governmentality” [88].

For example, while making judicial decisions about parole, probation, bail, and sentencing decisions, the criminal courts have usually been using intelligent algorithms in the criminal court as predictive justice practices; to “predict future dangerousness of an offender” [77] and determine an accused person’s risk of recidivism that “typically provide

background information on offenders to sentencing courts” [16].

Most notably, the American Law Institute recommends a broader use of risk-assessment tools in the most recent version of their highly influential Model Penal Code (see Ref. [94]). So evidently, AI algorithms are being enthusiastically chosen as a “potent, pervasive, unstoppable force” [80] looking forward to replacing obsolete bail and sentencing practices, and modernizing judicial practices, even in defiance of the fact that such systems are just about finding correlation or patterns in a vast trove of data and are not without risk (see e.g., [102]).

So far as the ambit of this paper goes on, it is pertinent to mention that the proponents of AI solutions particularly posit that by “predicting complex individual behaviors” (see Ref. [10]), the predictive algorithms can make analyze data in a more accurate and less biased way than humans can do. Their propositions also reveal that the predictive algorithmic outputs have the potentials to minimize intrinsic human biases and provide other welcome benefits, such as consistency, neutrality, and objectivity toward the evaluation of the information regarding an offender.

Conversely, there also arise deep concerns regarding the court’s use of bias and discriminatory data (demographic and socio-economic variables) as input in the currently used predictive AI system. Likewise, since they particularly purport to produce biased and discriminatory outcomes, they become adversely impactful on individuals’ rights, adjudicatory values,

¹ For this paper, artificial intelligence (AI) includes automated computer programs capable of helping or replacing the traditional judicial decision-making methods (e.g., algorithms used in predictive analytics).

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and judicial integrity as a whole. Consequently, unwanted consequences like algorithmic bias, racial discrimination, and mass incarceration may result from increased reliance on such predictive algorithms in a given criminal justice system (see Refs. [20, 106]); which makes such automated risk-scoring systems “constitutionally, technically, and morally troubling” [94].

Hence, it appropriately necessitates exploring and examining whether, how, and why the criminal courts’ AI system reinforces bias and discrimination in justice systems; and whether a given society should continue with such a current version of AI-assisted decision-making practices in the criminal court. As the paper’s primary focus delves into risk assessment tools’ algorithmic bias and disparate outcomes, arguments are also cemented as such that how these AI tools purportedly appear to have data-reflective bias-effects, and why they are allegedly reinforcing bias and discrimination in society while being used in criminal courts. It also concludes by suggesting tentative alternative approaches that may seem appropriate policy or systematic or institutional alternatives to the current predictive justice mechanisms.

2 AI systems in criminal courts and shifts in justice delivery mechanisms

As aforesaid, the rapid development in machine-learning algorithms and judicial data infrastructure have already demonstrated that AI systems possess vast potentials for being embraced in criminal courts to replace obsolete bail and sentencing practices, reduce recidivism risk, and modernize judicial practices. In fact, the law enforcement agencies’ use of predictive technology greatly influences the adoption of AI systems in criminal courts, which are constituent components of the core legal institutions.

Traditionally, judges in the criminal court practice sentencing criminals on written rules embedded in paper forms. However, such an approach remarkably causes a ‘vividly visualized shift’ mostly in the American jurisdiction and the few, e.g., Malaysia (see, e.g., [19]). The shift in criminal justice systems here means that “courts use predictive algorithms to be informed of prediction who is likely to fail to appear at their court hearing, and who is likely to re-offend at some point, in the future” [78]. Such practices are popularly known as predictive justice.² Such criminal justice predictive algorithms are also called ‘risk assessment tools’ or

‘evidenced-based methods’ (see Ref [56]). For example, the ‘COMPAS’³ is one of the well-known commercial products widely used within the US criminal justice system, where algorithms inform the bail, sentencing, and parole decisions for several decades across USA both at the federal and state levels (see Ref. [33]).

Basically, this is such a decision support software that uses algorithms to assess the potential risk of recidivism, which is calculated by taking into account both an interview with the defendants and information from their criminal history, including an accused person’s criminal record, age, gender, and answers to dozens of questions about their marital and family relationships, living situation, school, and work performance, and other risks factors; and then typically provide an estimate of an offender’s risk of recidivism, or risk of failing to appear when on bail, calculating an overall score that classifies an offender as low, medium, or high risky. It then helps judges make decisions regarding pre-trial detention, post-trial sentencing, and bond amounts or parole.⁴

Nevertheless, it “does not predict the specific likelihood that an individual offender will re-offend” [63]. Instead, the algorithm provides a risk prediction depending upon the comparison of a vast amount of data about an offender to a similar data group (use of proxy information). Hence, it is aptly put that “post-trial sentencing predictions forecast likely recidivism, and probability outcomes forecast likely probation violations” (see, e.g., [61, 62]). However, unlike criminal court applications, recidivism forecasts are not new in the USA. The criminal justice system of USA has been using personal characteristics, such as age, race, criminal history, school grades, geography, background and employment, and so on since the 1920s for assessing former inmates’ reoffending possibility and determining their special needs for social services like mental health or abuse treatment upon release (see Ref. [93]).

However, critics instead raise robust concerns over this system of codified justice and labeled the current actuarial assessments tools as “unreliable, controversial, and unconstitutional”; as being created and trained with biased data “produced through histories of exclusion and discrimination” (see, e.g., [5, 34], etc.). Therefore, the new trend of such quantification and standardization in criminal court systems appears to replicate biases and amplify discriminatory impacts that already exist in society, and thus, is doing

² The term ‘predictive’ in AI jargon is contextually linked to the possibility of predicting future results through inductive analysis, which identifies correlations between input and output data (see Ref. [89]). And the term ‘preventive justice’ was first used in the late eighteenth century and aimed at preventing future crime ([72], p. 1753). Over time, preventive justice schemes have been reinvigorated with risk assessment algorithms to predict recidivism risk.

³ The ‘COMPAS’ is created by the for-profit company Northpointe (which rebranded itself to ‘Equivant’ in January 2017). The recidivism risk scale of COMPAS has been in use since 2000 (see Ref. [31]).

⁴ As for predictive justice, there are nowadays reportedly ‘more than 200 risk assessment tools available in criminal justice and forensic psychiatry, which are widely used to inform sentencing, parole decisions, and post-release monitoring’ (see Ref. [63]).

otherwise than reducing judges' bias, discretion, and arbitrariness [67, 68].

Pertinently, Greengard [41] arguably apprehends these software predictions to be “more consequential” nowhere than in the criminal justice system. Most importantly, the famous ‘Loomis’ case⁵ still stands as the most prominent case in this respect. In Loomis, the trial judge gave Loomis a long sentence partially, because the defendant Loomis received the ‘high-risk score’ from COMPAS. Loomis challenged his sentence first in the state Supreme Court and then the Supreme Court of the USA, but the latter court ruled against Loomis (see Refs [66] [91]). Accordingly, it would not be verbose to opine that such an episode puts the criminal court's AI systems in an acid test that gains sufficient momentum aftermath the Loomis' verdict (see e.g., [55, 64, 90]).

3 Concerns over biases in risk assessment algorithms

The contemporary world has both exaltation and deference to technological solutions. However, since a plethora of legal and ethical issues over the use of algorithms in criminal courts have already arisen (see e.g., [82], this part of the paper particularly emphasizes the traditional issues like bias and discrimination, among varied implications of and deep concerns over the (so-called) popular and commercially available novel tools, and explores such an ongoing debate from different perspectives.

3.1 Transmission of human bias into the predictive machine

As a general rule, bias and discrimination in any form are explicitly barred in law except for the affirmative one. Accordingly, a complex issue in the ongoing debate regarding the current version of AI systems in use in the public sector throughout the world is around the concern over ‘biases’.⁶ The proponents for the risk assessment algorithm claim that by way of “predicting complex individual behaviors” (see Ref. [10]), they are intended to minimize ‘human biases’, as judges might abuse this unchecked power that could contribute to the unequal application of laws [41]. Minimizing ‘intrinsic biases in human decisions is notably

reasoned on the ground that the non-humans' decision-making process is less biased than humans.

As such, a growing tendency for deploying AI systems opts to justify in ways that predictive algorithms have neither hearts nor hands, nor emotion, whims, or choices or attitudes. Rather, they simply perform calculations to find correlation in the vast of data and provide “objective, neutral and reliable” outputs” [79]; that seemingly seems to be more “objective’ evaluation” [43], which could help judges uncover their own biases. Even when algorithms are used in the court's decision-making process, it will be easier to ascertain bias and discrimination (if any), or as it is claimed that, “at least, it should be” (see, e.g., [60]).

Therefore, it becomes a conspicuous proposition that vividly reveals a discourse that ‘more technology’ means potentially ‘more insights’. Since computers are not humans, machines do not have those problems that humans have. Therefore, it is most relevant to point out that where a criminal justice system involves human judges, their decisions may also have black box effects in some way, something like a judge just sitting in black robes, and thus, there is always a risk for bias decisions too. This is why, AI systems are expected to “vaporize biases and heuristics inherent in human judgment and reasoning, and thus, increase the legitimacy of the criminal justice agencies and confine sentencing to the pure scientific reasons” [108]. In fact, algorithmic assessments have already been employed in other public spaces than the criminal justice system, resulting in visible, eminent, and noteworthy outcomes.

Conversely, although the aforesaid arguments seem to be robust and challenging, it is trite to put that there are a considerable number of studies that rebutted those viewpoints on the premise that predictive assessment algorithms evidently appear as “re-purposed”, “unsustainable”, and “misguided” (see Ref. [71]) and, thus, make a call for a halt of the use of the in the justice sector. To clarify, this paper will not examine those perspectives that urge for a stop to using such AI tools in the public sector. Instead, it is trying to examine how the artificial intelligence systems currently deployed in criminal courts reinforce bias and discrimination in society.

First, it is commonplace to make recall that cognitive scientists have already discovered over 180 biases traceable in some AI systems (see Ref. [99]); which evidently supports the multiple forms of data bias. As human fallibility and inherent bias reflect through data, the biased data can produce biased results [58]. In consequence, the training data of AI algorithms mirror society as it exists, for better or worse and machine-learning models becomes truly “a representative of our biased world” [87, 100]. In addition, the so-called ‘algorithmic bias’ may also result from many careless mistakes in labeling data and oversights in data aggregation (see Lloyd and Hamilton [65]). For that reason, the data-supported system inescapably generates disparate

⁵ 881 N.W.2d 749 (Wis. 2016), cert. denied, 137S Ct. 2290 (2017); this was a Wisconsin Supreme Court case that was appealed to the United States Supreme Court but denied on June 26, 2017 (see Ref. [91]).

⁶ Algorithmic bias describes systematic and repeatable errors in a computer system that create unfair outcomes, such as privileging one arbitrary group of users over others. Bias can emerge due to many factors, e.g., the design of the algorithm or the way data is coded, collected, selected, or used to train the algorithm (see Ref. [59]).

impacts and prejudice against certain groups, such as protected, marginalized, or underrepresented groups.

With this in mind, it follows that the role of the data science lifecycle (data ingestion to data analysis) becomes technically skeptical and ethically problematic (see Ref [98]). As such, it is palpable in another sense that the ‘data minimization approach’ (see Refs. [67, 68]) may foil potential harm, since the risk assessment algorithm is still unable to correct human errors and biases. Hence, the claim that “with more technology comes more insights” appears to be not made from perspectives otherwise than the algorithms themselves, such as perspectives of individual rights, fairness, and prejudice (see Ref. [71]).

Second, it transpires that the scientific calculations or correlation of the risk assessment algorithm are often chosen for their “objective” evaluation (see Ref. [43]). However, considering their impacts, it becomes just opposite. Simply because such systems are data-driven, it would be mistaken to assume such techniques as objective (see Ref. [8]). Accordingly, Crawford [26] arguably disregarded the objectivity of data and datasets, and claimed that predictive algorithms present considerable risks for being a product of human design and having “hidden biases in both the collection and analytics”.

In fact, the risk assessment method yields probabilities, not certainties, and measures correlations, not causations. Furthermore, only finding correlations hinder from examining if they (correlations) are real or ridiculous [86]. On that account, an actuarial risk assessment seems not to be a neutral weapon to promote fairness and fight the implicit bias in the court proceeding (see Ref. [108]). That being the case, Dzindolet et al. [32] rightly argue that automated decisions in the age of AI are being rated more positively than neutral. Moreover, since the criminal justice system has already entered into the age of “datafication” [85]; human bias and arbitrariness will become also natural upshots: in ways that algorithms and databases are human artifacts fed with data, not exclusive of social, cultural, and economic circumstances. Machine bias thus inherits from humans as an inevitable consequence.

Third, questions may then follow on the roles of de-biasing: how it could be achieved, whether it would be dehumanization at the court’s decision-making loop. These are contemporary and legitimate questions in the present context. Studies find that nobody is expected to give any wholesome approbation to entirely eliminate human judges from the court’s bail or sentencing decision-making loop; at least, so far as the current version of algorithms in use is contextually under examination. Pointedly, although de-biasing through dehumanization may sometimes seem to be a pressing need, even if possible; this will desirably be “fairness through blindness” ([13], pp. 183–186). Pertinently, Završnik [108] also argued that taking resort to ‘machine bias’ placing no

human in the decision-making loop would be ‘a machine-based utopia’; where “judicial decisions would thereby be relegated to the experts of the computer science elite”.

Furthermore, although an explainable algorithm may provide a better idea about the data selection, data cleaning, data modeling, and data interpretation—a clear view about the way it reaches decisions, humans are still playing significant roles in the process of the prediction life cycle. As a result, the varied forms of biases in a risk-scoring system may be traced out in the very nature of the humans’ cognitive biases. Because of that, another question again arises whether society should prefer machine bias to human bias. The answer is, of course, in the negative too; at least, considering the aforesaid narrative. Therefore, it is necessarily fair to examine whether or how an increased quality of humans over computerized decisions should be the desired goal. However, it fairly stands outside the circumference of the paper (for more, see e.g., [27]).

3.2 Identifying varied biases in the court machine

Whereas bias exists in all data on earth, it undoubtedly occurs in diverse forms and natures. To date, researchers have already discovered in-cognitive biases encapsulated in datasets and algorithms currently in use, alongside numerous cognitive biases in predictive algorithms. As aforesaid, cognitive scientists identified over 180 biases in some AI systems (see Ref. [99]). The different forms of bias include sampling bias, representation bias, confirmation bias, technical bias, outcome bias, blind-spot bias, availability bias (heuristic), clustering illusions, and bandwagon effects, among others [108]. Whereas Chou et al. [18] recognized some major categories of bias, such as dataset bias, association bias, and interaction bias, Mehrabi et al. [70] categorized them with 23 separate sources of bias, additionally including measurement and linking bias, etc. Now, it requires exploring and examining in the criminal justice context.

Considering the Loomis court’s warning for judges proceeding with ‘caution’, a question may arise how human judges would defy an advanced and sophisticated technological tool’s burdensome prediction. Therefore, such cautions might be ineffectual eventually. Accordingly, judges’ sense of individualization faces risks for being treated as less reliable, less scientific, and less legitimate than the quantified prediction in courts. In the same way, judges’ mindsets might start assuming that “quantitative methods are superior to ordinary verbal reasoning” [76]. In our case, such an over-reliance tendency of judges on risk assessment algorithms could promote ‘automation bias’; giving control over to machines and thus, leading to accountability unpredictability. Even, over-reliance on such a first piece of information, risk prediction provides, may also generate ‘anchoring biases’ in the judges’ decision-making process

(see Ref. [20]); which has the potentials to easily change “a technological suggestion into a final, authoritative decision” [21].

Likewise, judges’ tendency to act upon only those outcomes that confirm their pre-perceptions may craft confirmation bias. In addition, dataset bias may reinforce inaccurate generalization, which is opposite to case individualization, a great virtue of the criminal justice system. Pertinently, it appears that problems in data sampling have similar impacts to representation bias for the former may compromise the representativeness of data subjects. Even, it is a fatal consequence that the insight generated out of analyzing aggregate data may be paradoxical when sub-groups are analyzed (see Ref. [74]). For example, those defendants who, as an individual or a group, constitute a less-proportionate contribution to the volume of sampling and training data may fall to the victim of miscontextualization of cases and the misapplication of laws. Therefore, a biased approach in data sampling and data scrubbing may categorize the proxy information of defendants as identifying one, such as minority or majority group; and eventually, advance exclusion bias by producing unethical, disparate, and inaccurate outcomes against the vulnerable group [75]. Technical bias also results from technological constraints or errors that outweigh particular groups without an underlying value [38].

Furthermore, social bias may also be unintentional and hidden—clandestine in any dataset used in the machine prediction (see Ref. [71]). Unintended bias against protected groups may also be derived from the lack of data diversity in training algorithms (see Lloyd and Hamilton [65]). Both the cases may “represent some hidden limitations and cautions” [22, 58]. Besides, algorithms in use may not always be facially biased and discriminatory, for no algorithms would intentionally be built as unconstitutional code in any jurisdiction. Instead, they may have an “implicit existence in the digital ecosystem” [104]. In such a case, it is difficult to determine who is responsible or accountable for mistakes the algorithm makes; akin to automation bias. Even, training data may well replicate unintended or historic bias. Therefore, it becomes lucid that only removing judges’ discretion from the judicial decision-making loop would be a double-edged sword; that is to say, as Završnik [109] opines, “it can reduce human bias, but it can also exacerbate past injustices or produce new ones”. As a result, such dangers conspicuously appear to be fatal in the high-level decision-making environment, which Israni [49] labels as both improper and inappropriate.

However, the COMPAS is not the only risk assessment instrument having such fundamental problems. AI algorithms are also being used in some other public and commercial space, such as predictive policing software ‘PredPol’, Azevea’s HunchLab; Amazon’s face recognition technology ‘Rekognition’, and NEC’s Neoface Live Face Recognition, etc. (see Refs. [23, 46]). A study published in 2019 reports

that a health care risk prediction algorithm used in the US hospitals to identify and help the patient with complex and extra health needs demonstrated racial bias: heavily favored white patient over black patients (see Ref. [73]). Even, in 2015, Amazon’s AI recruiting tools were found to be biased against women; the system “taught itself that male candidates were preferable” basing on past data [84].

Moreover, the impacts of the Chinese governments’ AI used for racially profiling Uighur minorities can be compared to that of the predictive algorithms of the USA’s courts. Even, it is pertinent to highlight the US Equal Employment Opportunity Commission’s online job interviews algorithm now under examination of algorithmic discrimination on the ground of racism and sexism. Hence, it becomes neatly clear how the criminal courts’ AI system evidently perpetuates and exacerbates existing human biases in society, similar to other public authorities, such as policing, finance, social welfare, etc.

3.3 As discriminative instruments

“Bias is a dimension of the decision-making itself, whereas discrimination describes the effects of a decision” [71]. Although laws forbid discrimination, making out a discriminatory effect is a daunting task; and determining indirect discrimination (understood as ‘disparate impact’ in the USA) is one of the most complex issues in the context of algorithmic decision-making. Being resulted in an open-ended standard and remaining hidden in data sets, it may not have a clear and easily applicable rule (see Ref. [9]). Borgesius [9] further mentions that as it disproportionately affects a protected group; it needs to prove with ‘statistical evidence’ that there is a *prima facie* discriminatory outcome for it may not always have a prior discriminatory intent.

For example, if predictive assessments made on the basis of illegitimate grounds—protected characteristics, such as race, sex (see Chander [17]), religion, or political or sexual orientation, or ethnicity, and so on, an opportunity to examine training data used in an offender’s recidivism risk assessment could reveal such direct discriminatory outcomes (disparate treatment). However, when a risk assessment algorithm trained on data that are not facially biased, an unfair and problematic decision may nevertheless be made in an unexplainable way that could reveal hidden discrimination (e.g., overrepresentativeness or unrepresentativeness, etc.), even that may remain “outside the scope of most non-discrimination laws” [9]. Copyrights, trade secrets, privacy, unexplainability, and the like usually support ‘covert discrimination’ in AI systems. Therefore, it becomes a difficulty to discover hidden or indirect discrimination of AI systems in use.

There exists a belief that risk-assessment tools will be catalysts to curb historical discrimination entrenched in the criminal justice system, dispel disparities, and promote

consistency in the construction of data when judicial decisions are made. As a result, these predictive tools appear to some quarters as a necessity, no longer a mere convenience (see Ref. [44]). The AI solution indeed offers numerous promises for judicial modernization, but it deserves not to be deployed in the court precinct so long as having critical legal and ethical problems associated with them. For example, the COMPAS's higher false-positive rate is discriminatory against the black population; and thus, it illegally poses as a discriminative instrument prejudicial to predictive parity.

Opacity and unexplainability in predictive algorithms inherently precludes bias checks (see Ref. [15]). However, exhuming systematic or intersectional discrimination necessitates analyzing the algorithm and the context in which it was analyzed. For example, when a black woman is discriminated against, it may not always be wise to conclude that it is discrimination 'against woman'; since she is prejudiced not only for being a woman (see Ref. [101]). In view of this, the rule of fairness warrants examining training data, code, and 'weight' attached to them. Whereas discrimination ousts fairness in decision-making, a modern form of risk prediction could create implicit and harmful discrimination as a by-product [50]. Therefore, without explainability, unintentional biases in datasets may be undetected. Hence, it can be said that an algorithm should fight such unfairness, not allowing discrimination against protected groups, like religious minorities, racial or ethnic groups, etc.

It is also crucial to examine who develops and deploys these systems and decides about data ingestion and data training. Interestingly, as the male population dominates AI research domains to date, women represent only 12% of them—"a worrying statistic for a field supposedly reshaping society" [92].⁷ Additionally, more data are available on men than women in the big data world (see Ref. [14]). On the other hand, the high-tech sector tends to "overrepresent white people and underrepresent black and Latinx people" (see Ref. [45]).

Accordingly, it becomes an algorithmic trend that risk assessment algorithms favor the overrepresented group by default. In effect, such representation bias creates a disparate impact on the court's fairness and legitimacy in decisions. As a result, the current version of AI systems in use in court premises is facially understood to reinforce both forms of discrimination (direct and indirect) in society. That is why, perhaps, Perez [14] artfully articulated in her book "Invisible Women" that due to the lack of big data on women, bias and discrimination are installed into our system: a system that treats men by default.

In some cases, such algorithmization itself was challenged in court on the allegation of producing discrimination and undue consequences in the public space other than courts. For example, the District Court of The Hague nullified the System Risk Indication (SyRI): an algorithm system, used by the Dutch social affairs ministry since 2014, to create 'the risk profile' and determine the 'likelihood' of welfare fraud (see Ref. [1]). This COMPUS-like-system also works with personal data ranging from income, pensions, employment to education, houses, and unemployment, etc. The court invalidated the system on the ground of "secrecy surrounding SyRI"; meaning that neither the court nor individuals can ascertain the decision-making process, which can lead to unexpected biases and indirect discrimination [39]. Similarly, an Italian court ruled out a rider-ranking algorithm (Deliveroo) for discriminatory assessment—not considering riders' legitimate reasons for ride cancellation while determining their reliability (see Ref. [1]).

However, one might argue that shifting judicial responsibility to predictive technology may better serve in removing human bias from the decision-making loop. Therefore, the idea of abdicating the responsibility for sentencing to computers can be justified, because they are "less biased than even a well-reasoned judge" (see Ref. [49]). However, it is also to be critically considered what would happen if any judicial system requires judges to be assisted with them. As stated hereinbefore, such a standardized requirement would make the court have 'no choice but to abdicate' a part of its responsibility to a hidden decision-making process. It may reinforce systematic discrimination in the justice system. Therefore, it should not be permissible in high-stakes decision-making, such as criminal courts.

To date, a silent shift of responsibility from public authorities toward private companies (or computer elites) is yet to prove that predictive AI systems better do so. Conversely, it is evident that such a shift seems to be highly problematic from the perspective of public authorities' demand transparency, responsibility, accountability, and fairness. Besides, a discriminatory AI system is not tolerable in the justice system, simply because it has the potentials to violate human rights and undermine public trust in such a system [105]. Hence, it is widely claimed that shifting the sentencing responsibility to a computer does not necessarily eliminate bias; rather, it reinforces and often compounds it by way of contributing to the spread of disparate impacts and discrimination in society.

3.4 Impacts of accuracy level

No doubt, there is a close link between inaccuracy and the principle of fairness and non-discrimination (see, e.g., [12, 110]). Although predictive AI systems are statistically believed to increase predictability in courtroom outcome,

⁷ At Google, only 21%, and at Facebook, only 22% of technical roles are filled by women. This estimate came from tallying the numbers of men and women who had contributed work at three top machine learning conferences in 2017 (see Ref. [92]).

lack of accuracy causes prejudice and produces discriminatory effects. In 2016, a ProPublica study⁸ had a close look and analyzed COMPAS assessments for more than 7000 arrestees, and published an investigation⁹ that claims that the algorithm was biased against black defendants (see Refs. [81, 107]). Conversely, as it evidences that, whites who became repeat offenders were disproportionately likely to be misclassified as ‘low risk’. It is also found (as COMPAS predicts) that blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend, and whites are much more likely than blacks to be labeled lower risk but go on to commit other crimes. It also appeared that “only 20% of the people predicted to commit violent crimes actually went on to do so” (see Ref. [51]).

However, in rejoinder, Northpointe argued that their test is unbiased, because it is equally predictive for black and white defendants, and the likelihood of recidivism among high-risk offenders is the same regardless of race (see, e.g., [25, 28, 31]). As they also asserted that it is rather ‘informative’ (60% accuracy is better than a coin flip), ProPublica again re-examined and analyzed the data from the perspective of the defendants, keeping the company’s criticisms in mind, and finally, stand by their conclusions; and also rebutted the rebuttal arguing that “since the whole point of due process is accuracy, the idea to live with a 40% inaccurate result is a mind-boggling way to think about accuracy” [51].

Inaccuracy issues are also unearthed regarding similar algorithms. For example, researchers from MIT and Microsoft conducted the Gender Shades Audit, 2018 and published a groundbreaking study called Gender Shades on the gender and racial biases embedded in commercial face recognition systems [11]. Their audit reveals that face recognition is more accurate for white than black people. Hence, the current accuracy level of the public algorithm shows default flows that undermine some ethical questions and the notion of fairness (see Ref. [24]).

In fact, the impacts of inaccuracy rate of the public sector’s AI apparently appear to be too essential to consider shallowly, for it could result in a death sentence based on the prediction of defendants’ dangerousness. Simonite [92] opines that the situation created by such practices is “much messier than in the hypothetical world of such studies”. For example, it cannot be gainsaid that as a single jurisdiction, USA locks up more prisoners than any other country; and the black population makes up a disproportionate amount of them (see Ref. [52]); which, in essence,

⁸ ProPublica, a nonprofit organization, aiming to produce investigative journalism in the public interest, looked at risk scores assigned to over 7000 people in Broward County, Florida, and checked to see how many were charged over the next 2 years, and found that COMPAS tool was ‘biased against blacks’. Such findings provoked a public debate about the problems of automating government systems [57].

⁹ However, a rejoinder was also made to this investigation report (see Ref. [37]).

reflects the US criminal justice system’s saga of disparate treatment. The inaccuracy rate of the AI systems currently in use in the criminal courts severely appears to be materials, as well.

As aforesaid, to argue in another sense, since standardized machine outcomes remain influential for judges “although they currently don’t always, or are not strictly obliged to, follow those outcomes” (see Ref. [54]). Consequently, it can be argued that “whereas risk recidivism algorithms like COMPAS have a 70% accuracy rate, how is it fair for a justice system that advertises as innocent until proven guilty to tolerate such an error rate” (see Ref. [48]).

3.5 Quantified disparity

As aforesaid, judges may likely estimate quantitative methods superior to ordinary verbal reasoning, and likewise, it is really challenging and unusual for them (judges) to defy algorithmic recommendations. Being “more cumbersome” upon judges to use these assessments as such, they (judges) may feel internal pressure coming from within the judicial system, which will, in return, push them to cope up with the predicted outcome [95].

Beyond this quantified pressure in criminal courts, risk assessment outcomes may greatly influence the human judicial mind to promote cognitive biases leading to disparate decisions, which itself is a product of the pre-designed tools developed by only the non-judicial bodies. Besides, the widespread endorsement of these sentencing tools makes grow up a sense in the mind of judges that the tools are somehow reliable. Here grow psychological biases by way of quantification which, in turn, tend to support codified (in) justice “at the expense of equitable justice free from bias and discrimination” [85].

As discussed above, quantification in the criminal court may produce quality representative values independent of humans; and bring in uniformity and standardization above judges’ discretion (see, e.g., [85], pp. 67, 68). Giving over-reliance on standardization through quantification, there may arise a potential proposal of ‘complete’ dehumanization in the court’s decision-making process. However, persuasive evidence is rare on this point to date; at least so far as judicial systems are in our context. Consider a traditional discourse, computers may know many things and be intelligent, but they are not wise.¹⁰ Machine learning and AI algorithms themselves can learn by experience, thus learning our biases too. Therefore, the use of smart computation and dehumanization at the premise of the court’s decision-making loop may not have the intended results. Instead, it would act as

¹⁰ An interesting difference between intelligence and wisdom is knowledge is knowing that a tomato is a fruit; wisdom is not putting it in a fruit salad (Miles Klinton’s witticism).

an agent of disparity. Therefore, it is not skeptical anymore that complete reliance on risk prediction may not be a viable solution for an extremely sensitive context like the justice system at all.

In addition, recidivism assessment in sentencing does not sufficiently incorporate causation; or does not have any ability to individualize a case what a human has. Instead, risk assessment algorithms often consider more than 100 personal characteristics—such as socio-economic status, age, sex, geography, background, employment status, and neighborhood crime (see Ref. [104]). Accordingly, considering the aforementioned prohibited criteria and not considering causation and individualization of cases, such algorithmic machines would drive judges “to make disparities worse” [92]. Therefore, in line with Professor Starr’s [94] comment, it is all likelihood that the risk assessment with problematic variables creates disparate treatment concerns; that reflect the “penological interests of the state weakly if at all”.

4 The way forward

No doubt, the impacts of new technology are not always a happy one. As aforementioned, critics have underscored AI systems’ role in reinforcing existing bias and discrimination in the context of the criminal court system. However, in addition to that, the positive effects and promises of AI systems should not be forgotten. Therefore, striking a balance between the AI systems’ benefits for the administration of justice and the harms of disrespecting the core principles of criminal justice (see Ref. [63]) necessitates finding out the possible way out.

The challenges the criminal justice and legal communities are facing in the adoption of AI methods may be mitigated by taking recourse to the two potential paths: ‘either blindly allow the march of technology to go forward’, or ‘create a moratorium’ on the use of opaque AI in criminal justice risk assessment until being allowed for a meaningful examination of these tools (see Ref. [96]). In effects, preferring a pause and creating a preventative moratorium would facilitate courts in taking more time in determining how AI-driven risk assessments should be utilized in courts; and give policymakers, tech businesses, educational and advocacy organizations the breathing space to teach the users how to handle the AI solution tools being used in court effectively.

Although the EU legal framework requires companies to ensure clarity in the decision-making process (Goodman and Flaxman [40]), an idea of ‘full transparency’ (openness of data, code, logic with explainability, and interpretability) may be promoted. Apart from “checking out data quality”, an instrument of “impact assessment” might be carried out (see Ref. [36]) in the lights of the established judicial principles and legal rules, such as non-discrimination, due

diligence, procedural fairness, and privacy, etc. Instead of quantification in courts, Žliobaite [111] suggests for “quantifying discrimination that could help promote equity in application”.

In addition, the potential risks with risk assessments are often associated with ‘purposes’; the principle of purpose limitation (a principle of data protection law) has close reference to it as it inhibits collected data from being used for purposes other than the original purpose (see Ref. [42]). Pertinently, the criminal court’s algorithms come from computer science, rather than law. Nevertheless, despite having disruptive and transformative effects, they are chosen to produce a prediction relevant to the criminal justice system (see Ref. [20]). For example, the COMPAS system specifically was created, “not for use in sentencing, but rather to aid probation officers in determining which defendants would succeed in specific treatment types” [34]. Since such a risk assessment algorithm determines the statistical probabilities and has already been reported with having serious and unsettled issues, including “bias in its dirty data sets and statistical models”, this computational method appears to be antithesis to the core foundation of the justice system [47]. Accordingly, there is still large room for reconsideration and reexamination in the context of court system when those algorithms are re-purposed for which they were not originally designed or intended for. That is why; more attention requires to be drawn toward how those datasets are constructed and trained, and such models should be chosen with extra-caution through carefully identifying the criminal court needs and challenges [83]; “simply because a debiasing approach cannot be an afterthought” (see Lloyd and Hamilton [65]).

Moreover, since risk assessment algorithms often disguise “overt discrimination” (see Starr [94]), an installation of unexplainable computational logic into the judicial system might promote a “crabbed and inhumane vision” [76]; although it may appear as a standardized, scientific one. Besides, only building better algorithms is not a solution too. It is also not enough to ask whether algorithms predict correctly. It needs to look at an objective evaluation method with a sufficient level of fairness, transparency, and ethical values. It requires being able to say what should be, alongside just what it is. We then need to ask if it makes society better or worse. Therefore, instead of risk-scoring of an offender, a ‘risk mitigation’ approach is recommended with a view to understanding the social, structural, and psychological drivers of crime [6]. On this premise, Dignum [30] artfully argues that “removing the algorithmic black box does not suffice, and thus, will not eliminate the bias”. She instead suggests installing ethical principles and human values at the core of system design by default and calls for a mind-shift of researchers and developers toward a new

generation of algorithms, “which can turn the machine learning into valuable learning” (see Ref. [29]).

For diagnosing discrimination in AI systems, auditing on the actual decision-making systems, instead of mere lab simulation, could also be a viable option. A regulatory framework can help achieve it. If truth be told, every state has an obligation to explore alternative policies.¹¹ For example, ‘certification’ may be a viable option. In addition, applying the principle-4 of the CEPEJ Ethical Charter (transparency, impartiality, and fairness),¹² the judiciary is needful to have the choice of what type of system would best meet the institutional requirements. Even, building an algorithm completely, facially neutral would not suffice; rather, it would also affirmatively be trained against the racism of the world (see Ref. [48]), or it includes greater data diversity and inclusion in design and training models. As a meanwhile option, the risk assessment should be “limited to probation hearings or pre-trial release; and not to be in sentencing at all” [34]. Most importantly, judges should be a bias check on a tool itself designed to correct judges’ biases. Pertinently, the EFF’s proposal for imposing restrictions on the use of scoring systems can be considered too (see Ref. [103]). Moreover, the Universal Guidelines for Artificial Intelligence can also guide the use of algorithms in the pre-trial risk context [97].¹³

Furthermore, if judges have control over the court’s decision-making process, they themselves could potentially check bias on an algorithmic tool. The human nature of the court system suggests that criminal courts must primarily be run by human beings. To illustrate, if you understand that courts are humans, it would signify a human ability filled with intelligence, experience, legal knowledge, decency, fair-mindedness, patience, and balanced good judgment is utilized to evaluate complex evidence and apply nuanced legal reasoning to cases past and present with competing possible outcomes [3]. If you again think that courts are

humane, it then denotes human qualities that involve human reasoning, thoughts, values, emotion and human institutions, and the like (see Ref. [3]).

Consequently, since the process of datafication is alleged to threaten “the fabric of human agency” [4], it can be argued in an orthodox sense that a de-biasing approach through complete dehumanization at the court’s decision-making loop or complete reliance on risk prediction might promote “blindness”. Thus, human control over bail and sentencing decisions must exist in the criminal court systems even if more reliability, transparency, or accuracy in AI systems is achieved in the future. No non-human component can supplant the human sense of justice for an individual. Instead, such predictive AI systems can only be allowed to aid human judges to enhance court efficiency and cultivate the promises of the sophisticated AI enabled technology in other appropriate cases,¹⁴ but not to pursue quantification and standardization above judges’ discretion.

5 Conclusion

The discussion hereinbefore underscores that the version of the proprietary artificial intelligence systems currently in use of the criminal courts appears by nature to be reinforcing bias and discrimination in society. As a result, complex issues and challenges have already resulted in the criminal justice system due to the proliferation and rampant adoption of predictive algorithms. Since the domain of the criminal justice system is yet to transform into a hotspot for algorithms and data analytics, AI systems as the “least worst” option [69] could inform criminal courts until ethical and legal values are incorporated at the core of the system design by default; but complete dehumanization is discouraged all the time so far. No justice system thus should allow the counter-productive cool-blooded unchecked machines to blindly make the core institutions of criminal justice dive into algorithmic mania. Accordingly, this impetus appeals to the payers in this field for making a mind-shift toward building transparent and explainable AI systems with heterogeneous data; which would produce ontogenetic, purposeful, and valuable learning for criminal courts. Judges also should play a lead role as a bias checker alongside the machine insights. A pressing need further requires introducing an institutional and legal framework that could help better the justice system with the fairness, and without bias and discrimination.

¹¹ A recent example includes that a new bill is introduced in the senate of the United States in April of 2019 [7] that represents one of the first major efforts to regulate AI [2]. The new bill would require big companies to audit their machine-learning systems for bias and discrimination in an ‘impact assessment’, and take corrective action in a timely manner if such issues were identified. Notably, the US is not alone in this endeavor; jurisdictions like UK, France, Australia, and others have all recently drafted or passed legislation to hold tech companies accountable for their algorithms (see Ref. [53]).

¹² The European Commission for the Efficiency of Justice (CEPEJ) of the Council of Europe has adopted the first European Ethical Charter on the use of artificial intelligence in judicial systems. This first European text sets out ethical principles relating to the use of artificial intelligence (AI) in judicial systems (see Ref. [35]).

¹³ The guideline sets forth 12 principles that are intended to guide the design, development, and deployment of AI, and frameworks for policy and legislation.

¹⁴ AI’s potential can address other specific challenges that criminal courts face, such as processing and managing digital data, information-sharing, improving case management, evidence management, cyber security, resource allocation, and language translation, etc. (see Ref. [83]).

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